

Tezpur University

Spring Semester End Examination, 2022

BT 201: Biology for Engineers

Marks: 60

Time: 3 hr.

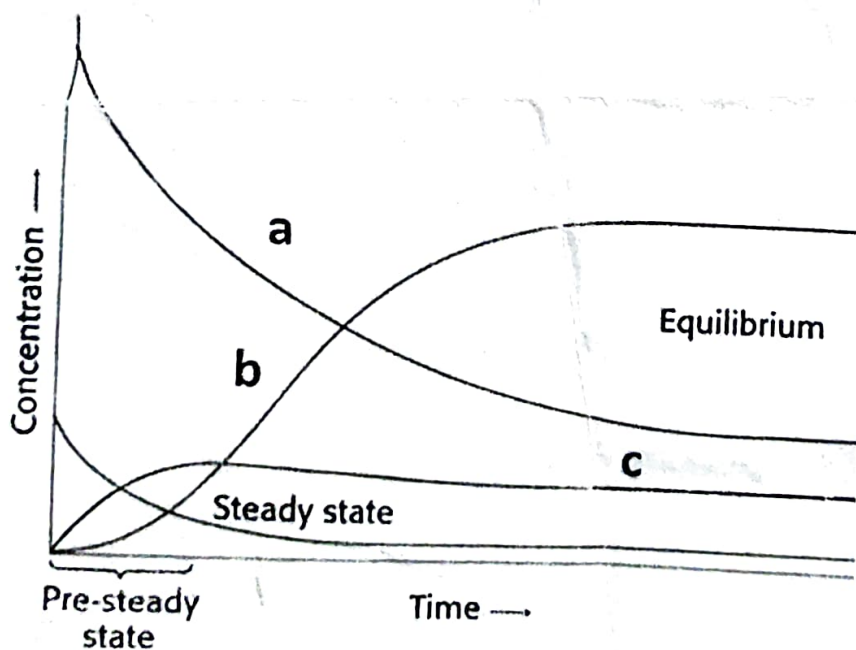
(Instructions: Answer Part A in answer book let provided and Part B in the question paper itself)

Part A (26 Marks)

Q1. Answer the following four questions:

[5×4=20]

- Draw a leveled diagram of a cell in prophase and late anaphase during mitotic cell division.
- Show the genotypic and phenotypic ratio of F₂ generation in a monohybrid cross experiment with an example.
- The graph below depicts the steady state assumption in enzyme kinetics. What does the label a, b and c represent? What is K_m in enzyme kinetics?



- Draw a labeled diagram showing growth curve of micro-organisms with different stages.

Q2. Answer the following two questions

[3X2=6]

- a) Match the following in a way that all group A word get connected to most appropriate group B items once only (multiple linking will be considered as wrong answer).

Group A	Group B
Branched	Lipid
Linear	Starch
Water binding	Amylopectin
Hydrophobic	Amylose
Sweet	Glucose
Reducing	Lactose

- b) Identify the wrong label in the below diagram and write the correct name in place after drawing the diagram on answer sheet:

